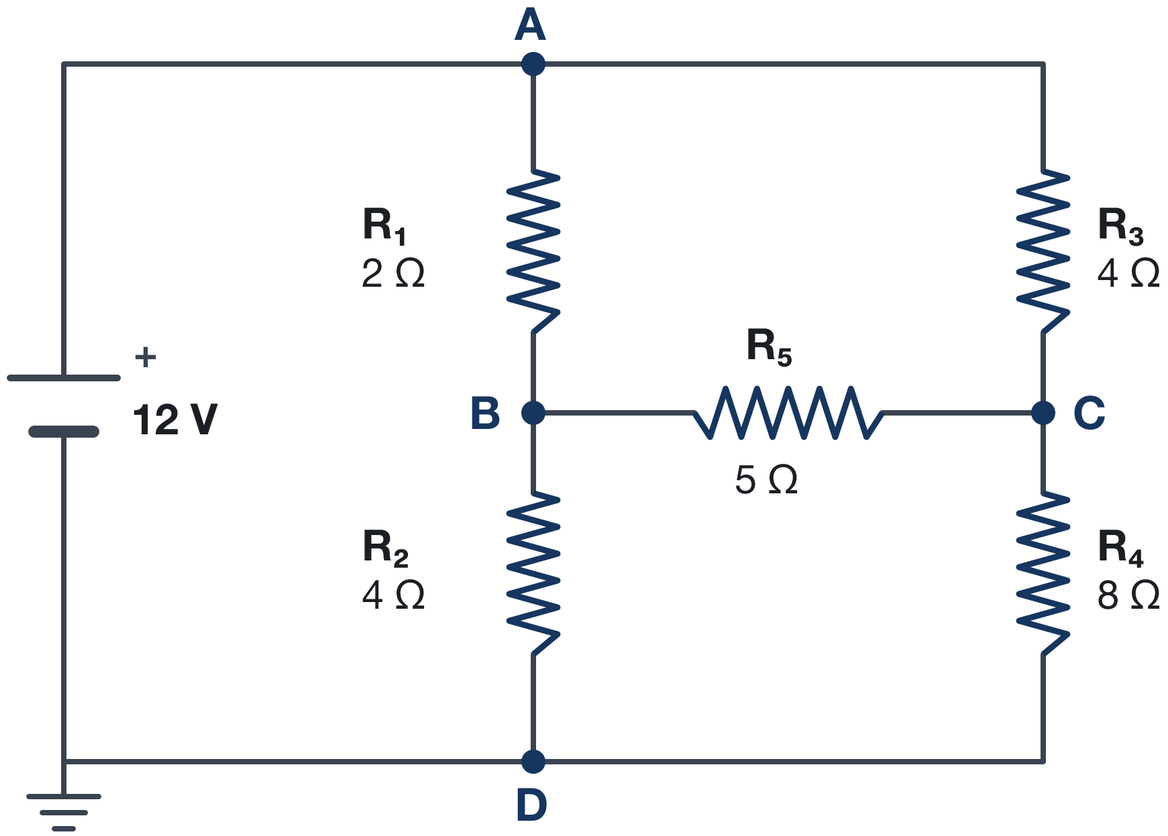
Wheatstone Bridge: Worked Example



Find the current through R_5.
Ideal battery and wires; D is ground.

1. Take `V_D = 0`, so `V_A = 12 V`.
2. Left divider: `V_B = 12 · 4/(2+4) = 8 V`
3. Right divider: `V_C = 12 · 8/(4+8) = 8 V`
4. `I₅ = (V_B − V_C)/R₅`

Balanced: `R₁/R₂ = R₃/R₄`, so $\mathbf{I_5 = 0}$